5.3 Metric Spaces

Definition 5.3.1 (Metric Space)
*Let X be a set and $d : X \times X \to \mathbb{R}_{\geq 0}$ a function. We say that (X, d) is a **metric space** if the following properties hold*

a) $d(x, y) = 0 \iff x = y$ for all $x, y \in X$
b) $d(x, y) = d(y, x)$
c) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$

*Given $x \in X$ and $\varepsilon > 0$ we define the **open ball** at x to be*

$$B(x; \varepsilon) := \{y \in X \mid d(x, y) < \varepsilon\}$$

Proposition 5.3.2
Let (X, d) be a metric space. Then we say that a subset $U \subset X$ is open if

$$\forall x \in X \exists \varepsilon > 0 \text{ s.t. } B(x; \varepsilon) \subset U$$

The open sets form a topology with open balls forming a base.

For each $x \in X$ a local base consists of open balls of the form $B(x; \varepsilon)$. Further we may also consider the countable local base

$$\mathcal{B}_x := \left\{ B\left(x; \frac{1}{n}\right) \;\middle|\; n \in \mathbb{N} \right\}$$

*In particular every metric space is **first countable**.*

Proposition 5.3.3
A metric space is Hausdorff.

Proposition 5.3.4 (Convergent Sequence)
Let (X, d) be a metric space and (x_n) a sequence in X and $x \in X$. Then the following are equivalent

a) For every $\varepsilon > 0$ there exists N such that $n \geq N \implies d(x_n, x) < \varepsilon$
b) The function

$$\begin{aligned} \mathbb{N} \cup \{\infty\} &\to X \\ n &\to x_n \\ \infty &\to x \end{aligned}$$

is continuous at ∞ with respect to the topology on $\mathbb{N} \cup \{\infty\}$ given by open sets of the form $\{N, N+1, \cdots\} \cup \{\infty\}$ and arbitrary subsets of $\mathbb{N}$. Note this function is automatically continuous at every n.

*In this case we say that $x_n \to x$ is a **convergent sequence**.*

Proposition 5.3.5 (Continuous at a Point)
Let $f : (X, d) \to (Y, d')$ be a map of metric spaces. Then for $x \in X$ and $y := f(x)$ the following are equivalent

a) For all $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d(y, x) < \delta \implies d'(f(y), f(x)) < \varepsilon$$

b) $f : X \to Y$ is continuous at x in the sense of topological spaces (4.1.25)
c) For every sequence $x_n \to x$ we have $f(x_n) \to f(x)$

Proposition 5.3.6 (Continuous Criteria)
Let $f : X \to Y$ be a map of metric spaces. Then the following are equivalent

a) f is continuous at every $x \in X$ (5.3.5)
b) f is continuous in the topological sense (4.1.27)
c) For every convergent sequence $x_n \to x$ we have $f(x_n) \to f(x)$

Definition 5.3.7 (Product Metric Space)
Let (X_i, d_i) be metric spaces for $i = 1 \cdots n$. Define the metric space on $X_1 \times \cdots \times X_n$ by

$$d((x_1, \cdots, x_n), (y_1, \cdots, y_n)) = \max\left(d_1(x_1, y_1), \cdots, d_{n(x_n, y_n)}\right)$$

Then the topology induced on (X, d) coincides with the product topology.

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5.3.1 Completeness

Definition 5.3.8 (Cauchy Sequence)
*Let (X,d) be a metric space. A sequence (x_n) in X is said to be **cauchy** if*

$$\forall \varepsilon \exists N \text{ s.t. } n,m \geq N \implies d(x_n,x_m) < \varepsilon$$

Definition 5.3.9
*A metric space (X,d) is said to be **complete** if every cauchy sequence is convergent.*

Proposition 5.3.10
Let (x_n) be a Cauchy sequence in a metric space (X,d) with $x \in X$ a cluster point (equivalently, subsequential limit point). Then $x_n \to x$.

Proposition 5.3.11
Let (X,d) be a complete metric space. A closed subset $Y \subset X$ is complete iff it is closed in X.

Proposition 5.3.12
Let (X_i,d_i) be complete metric spaces for $i = 1 \cdots n$. Then the product metric space is $(X_1 \times \cdots \times X_n, d)$ is complete.

Corollary 5.3.13
The metric space $\mathbb{R}^k$ induced by the product metric is complete.

A subset $X \subset \mathbb{R}^k$ is complete if and only if it is closed.

Proof. This is simply (5.3.12) and (5.3.11). ∎

5.3.2 Compactness

Definition 5.3.14
*We say a metric space (X,d) is **separable** if there exists a countable dense subset X_0.*

Proposition 5.3.15
Suppose X is a separable metric space. Then it is second countable.

Proof. We may show that the family

$$\mathcal{B} \coloneqq \left\{ B\left(x_0; \frac{1}{n}\right) \;\middle|\; x_0 \in X_0, n \in \mathbb{N} \right\}$$

is a countable base. ∎

Proposition 5.3.16 (Precompact)
Let (X,d) be a metric space. The following are equivalent

a) For all $\varepsilon > 0$ there exists a finite covering $X = U_1 \cup \cdots \cup U_n$ such that $\operatorname{diam}(U_i) < \varepsilon$
b) For all $\varepsilon > 0$ there exists a finite set of points $x_1, \cdots, x_n$ such that $X = \bigcup_{i=1}^n B(x_i;\varepsilon)$
c) Every sequence has a cauchy subsequence.

*In this case we say X is **precompact** or **totally bounded**.*

Proof. $a) \iff b)$ is straightforward. For $c) \implies b)$ suppose X did not satisfy this property, then we could inductively choose a sequence x_n such that

$$x_n \in X \setminus \bigcup_{i=1}^{n-1} B(x_i;\varepsilon)$$

Then evidently $d(x_n, x_{n+1}) \geq \varepsilon$ for all n and it can have no cauchy subsequence.

For $b) \implies c)$ Let (x_n) be a sequence. If $\{x_n\}$ is finite then we are done. Otherwise assuming it is infinite, we construct a decreasing sequence of sets

$$A_1 \supset \cdots \supset A_k \supset \cdots$$

such that $\operatorname{diam}(A_k) < \frac{1}{k}$ and $A_k \cap \{x_n\}$ is infinite. For suppose $A_1, \cdots, A_k$ are constructed, then evidently A_k itself satisfies $b)$ so we may find $A_k = \bigcup_{i=1}^m B\left(x_i; \frac{1}{k+1}\right)$. For at least one i we must have $B\left(x_i; \frac{1}{k+1}\right) \cap A_k \cap \{x_n\}$ is infinite. Therefore we may define $A_{k+1} \coloneqq B\left(x_i; \frac{1}{k+1}\right) \cap A_k$. Finally choosing an increasing sequence n_k such that $x_{n_k} \in A_k$ yields the required cauchy subsequence. ∎

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Corollary 5.3.17 (Compactness Criteria)
Let (X,d) be a metric space then the following are equivalent

a) X is compact
b) X is countably compact
c) X is sequentially compact
d) X is complete and precompact

Proof. $a), c) \implies b)$ This is (4.1.106).a)

$a), b) \implies c)$ This is (4.1.106).b)

$c) \implies d)$ By (5.3.10) every cauchy sequence is convergent. Therefore X is complete, and precompact by (5.3.16).

$d) \implies a)$ We show that (X,d) is sequentially compact and second countable, and the result follows from (4.1.106).c). Sequential compactness is a consequence of (5.3.16) and (5.3.10). To show that X is second countable we may for each n find a finite cover

$$X = \bigcup_{i=1}^{N_n} B\left(x_{ni}; \frac{1}{n}\right)$$

We claim that

$$\left\{B\left(x_{ni}; \frac{1}{n}\right)\right\}$$

is a base. For given $y \in X$ and a nbhd $U \in \mathcal{U}_y$ we have $B(y;\varepsilon) \subset U$ for some $\varepsilon > 0$. Choose n such that $\frac{1}{n} < \frac{\varepsilon}{2}$. Then for some $1 \leq i \leq N_n$, $y \in B\left(x_{ni}; \frac{1}{n}\right)$. We may show that $B\left(x_{ni}; \frac{1}{n}\right) \subset B(y;\varepsilon) \subset U$ which shows that the given collection is a base. ∎

5.3.3 Uniform Continuity

Definition 5.3.18 (Uniformly Continuous)
*Let (X,d) and (Y,d') be metric spaces and $f : X \to Y$ a function. We say that f is **uniformly continuous** if*

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } d(x,y) < \delta \implies d'(f(x), f(y)) < \varepsilon$$

Proposition 5.3.19 (Compact & Continuous $\implies$ Uniformly Continuous)
Let $f : X \to Y$ be a continuous map of metric spaces and suppose X is (sequentially) compact. Then f is uniformly continuous.

Proof 1.. Fix $\varepsilon > 0$ then for every $x \in X$ there exists δ_x such that $f(B(x;\delta_x)) \subseteq B(f(x);\varepsilon)$. Evidently $X = \bigcup_x B\left(x; \frac{\delta_x}{2}\right)$ whence there is a finite subcover associated to $x_1, \cdots, x_n$. Define

$$\delta := \min\left(\frac{\delta_{x_1}}{2}, \cdots, \frac{\delta_{x_n}}{2}\right)$$

Given $y \in X$ then $y \in B\left(x_i; \frac{\delta_{x_i}}{2}\right)$ for some $i = 1 \cdots n$. Furthermore suppose $d(y,z) < \delta$ then

$$d(x_i, z) \leq d(x_i, y) + d(y, z) < \delta_{x_i}$$

whence $y, z \in B\left(x_i; \delta_{x_i}\right)$. Therefore by construction $d(f(y), f(z)) < \varepsilon$, as required. ∎

Proof 2.. By (5.3.17) we know that X is sequentially compact. Suppose that f is not uniformly continuous, then there exists $\varepsilon > 0$ and sequences $\{x_n\}, \{y_n\}$ such that $d(x_n, y_n) < \frac{1}{n}$ and $d(f(x_n), f(y_n)) > \varepsilon$. By assumption there is a subsequence n_k such that $x_{n_k} \to x$ and $y_{n_k} \to y$. Evidently

$$d(x,y) \leq d\left(x, x_{n_k}\right) + d\left(x_{n_k}, y_{n_k}\right) + d\left(y_{n_k}, y\right) \to 0$$

whence $d(x,y) = 0$ and $x = y$. By (...) $f\left(x_{n_k}\right) \to f(x)$ and $f\left(y_{n_k}\right) \to f(x)$ which is a contradiction. ∎

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